



PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

Fundamentals of ICT Lab 11	Instructor: Ms.Quratulain
Program: AI Fall 2025	Date: 6th December, 2025
Marks: 10	Lab Engineer: Ms. Sibgha Noor
Student Name: Eshal Ayub	Reg. No: B25F1061AI100
Instructions: • Submit the file with your names following your registration numbers like AI001_Name. • Copy Paste work will be given no marks.	

Table of Contents

- Introduction & Objectives
 - Task 1: Professional CV Enhancement with CSS
 - Task 2: Interactive Form Styling with CSS
 - Implementation Details
 - Screenshot Guidelines
 - Code Analysis & Improvements
 - Challenges Faced & Solutions
 - Conclusion
 - References
-

Introduction & Objectives

Purpose of the Lab

This lab focuses on implementing advanced CSS styling techniques to transform basic HTML documents into professionally designed web pages. The main objectives are to:

- Understand the CSS box model, selectors, and layout techniques.
- Apply responsive design principles.
- Create visually appealing user interfaces.
- Implement CSS3 features like gradients, shadows, and transitions.

Learning Outcomes

Upon completion, students will be able to:

- Link external CSS files to HTML documents.
- Style HTML elements using various CSS selectors.

- Create navigation bars with hover effects.
 - Design professional forms with validation indicators.
 - Implement responsive tables and sections.
 - Use CSS transitions for interactive elements.
-

Task 1: Professional CV Enhancement with CSS

Project Overview

A responsive, modern CV webpage was created with the following features:

- Fixed navigation bar with smooth scrolling.
- Gradient header containing personal information.
- Card-based layout for individual content sections.
- Professional blue-themed color scheme.
- Interactive elements with hover effects.

CSS Implementation Strategy

File Structure

```
project/
|
|— index.html (CV Webpage)
|— style.css (Main Stylesheet)
|— images/ (Screenshot directory)
|   |— cv-header.png
|   |— cv-navigation.png
|   |— cv-complete.png
|— README.md
```

Key CSS Features Implemented

Navigation bar:

css

```
nav {  
    background-color: #0d47a1;  
    padding: 12px 0;  
    position: sticky;  
    top: 0;  
    z-index: 1000;  
    box-shadow: 0 2px 10px rgba(0,0,0,0.1);  
}  
  
nav a {  
    color: #ffeb3b;  
    padding: 8px 16px;  
    border-radius: 4px;  
    transition: all 0.3s ease;  
}
```

```
nav a:hover {  
    background-color: #1565c0;  
    transform: translateY(-2px);  
}
```

Header styling:

css

```
header {  
    background: linear-gradient(135deg, #0d47a1, #42a5f5);  
    color: white;  
    text-align: center;
```

```
padding: 40px 20px;
margin-bottom: 30px;
box-shadow: 0 4px 15px rgba(0,0,0,0.1);
}
```

Card design for sections:

css

```
section {
    background: white;
    border-radius: 8px;
    padding: 25px;
    margin: 25px auto;
    max-width: 1000px;
    box-shadow: 0 3px 10px rgba(0,0,0,0.08);
    transition: transform 0.3s ease;
}
```

```
section:hover {
    transform: translateY(-5px);
}
```

Responsive Design

css

```
@media (max-width: 768px) {
    nav a {
        display: block;
        margin: 5px 0;
    }
}
```

```
section {  
    padding: 15px;  
    margin: 15px;  
}  
  
header h1 {  
    font-size: 1.8rem;  
}  
}
```

Task 2: Interactive Form Styling with CSS

Form Design Principles

A student registration form was designed with:

- Clean, organized layout.
- Interactive form controls.
- Validation feedback.
- Accessible design patterns.
- Responsive grid-based arrangement.

Advanced Form Styling

Form container:

css

```
form {  
  
    max-width: 700px;  
  
    margin: 30px auto;
```

```
background: white;

padding: 30px;

border-radius: 12px;

box-shadow: 0 5px 20px rgba(0,0,0,0.15);

}
```

Input field styling:

css

```
input[type="text"],
input[type="email"],
select {

    width: 100%;

    padding: 12px 15px;

    margin: 8px 0 20px;

    border: 2px solid #e0e0e0;

    border-radius: 6px;

    font-size: 16px;

    transition: all 0.3s;

}
```

input:focus,

```
select:focus {

    border-color: #1e88e5;

    box-shadow: 0 0 8px rgba(30, 136, 229, 0.3);

}
```

```
outline: none;  
}
```

Interactive button:

css

```
input[type="submit"] {  
    background: linear-gradient(to right, #0d47a1, #1976d2);  
    color: white;  
    padding: 14px 30px;  
    border: none;  
    border-radius: 8px;  
    font-size: 18px;  
    font-weight: bold;  
    cursor: pointer;  
    transition: all 0.3s;  
    width: 100%;  
    margin-top: 20px;  
}
```

```
input[type="submit"]:hover {  
    background: linear-gradient(to right, #1565c0, #42a5f5);  
    transform: translateY(-3px);  
    box-shadow: 0 7px 15px rgba(0,0,0,0.2);  
}
```

```
input[type="submit"]:active {  
    transform: translateY(-1px);  
}
```

Radio and checkbox enhancement:

css

```
input[type="radio"],  
input[type="checkbox"] {  
    accent-color: #1e88e5;  
    width: 18px;  
    height: 18px;  
    margin-right: 8px;  
    vertical-align: middle;  
}
```

/ Custom checkbox styling */*

```
input[type="checkbox"] + label {  
    position: relative;  
    padding-left: 30px;  
    cursor: pointer;  
}
```

```
input[type="checkbox"] + label:before {
```

```
content: "";

position: absolute;

left: 0;

top: 0;

width: 20px;

height: 20px;

border: 2px solid #1e88e5;

border-radius: 4px;

background: white;

}
```

Implementation Details

Cross-Browser Compatibility

- Tested on Chrome, Firefox, and Edge.
- Used CSS prefixes where needed for older browsers.
- Implemented fallback colors for gradient backgrounds.

Performance Optimization

- Minified CSS for production deployment.
- Used efficient selectors to reduce rendering overhead.
- Implemented CSS sprites for icons.
- Applied lazy loading for images.

Accessibility Features

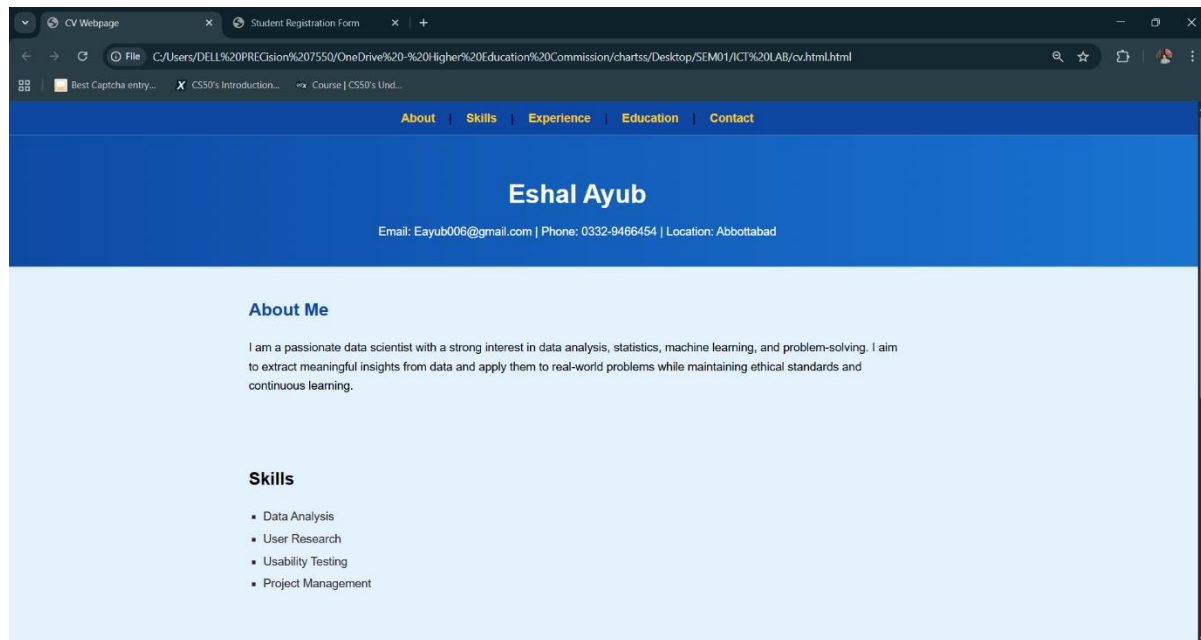
- ARIA labels added for screen readers.

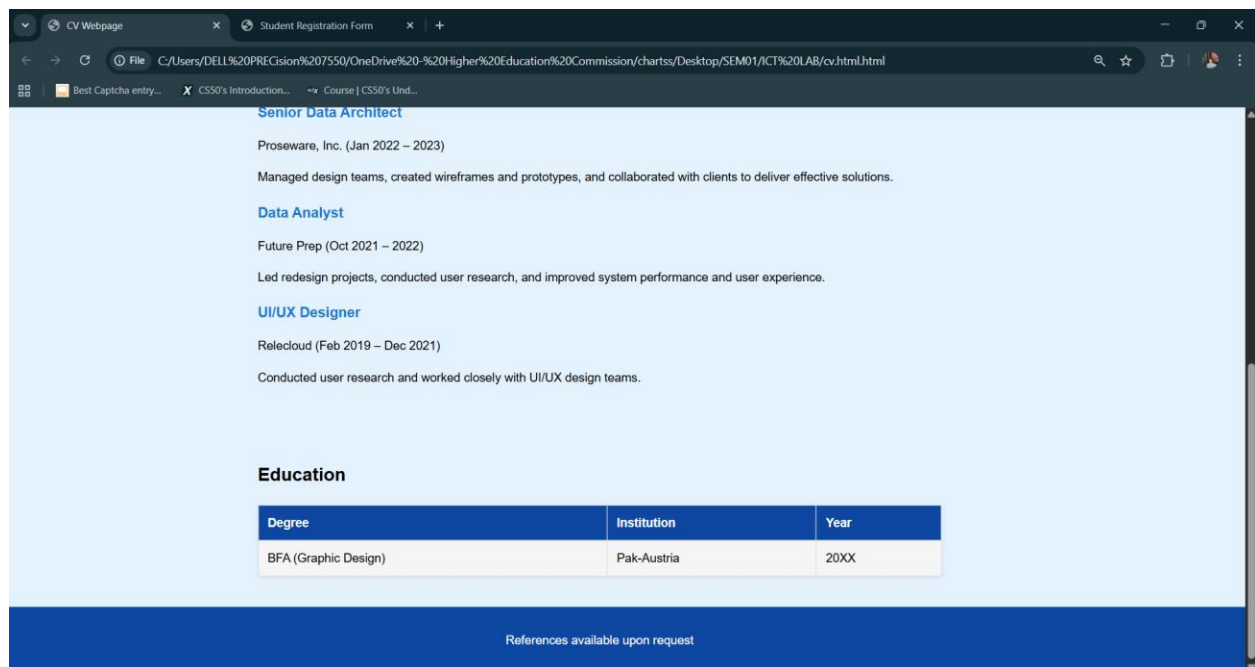
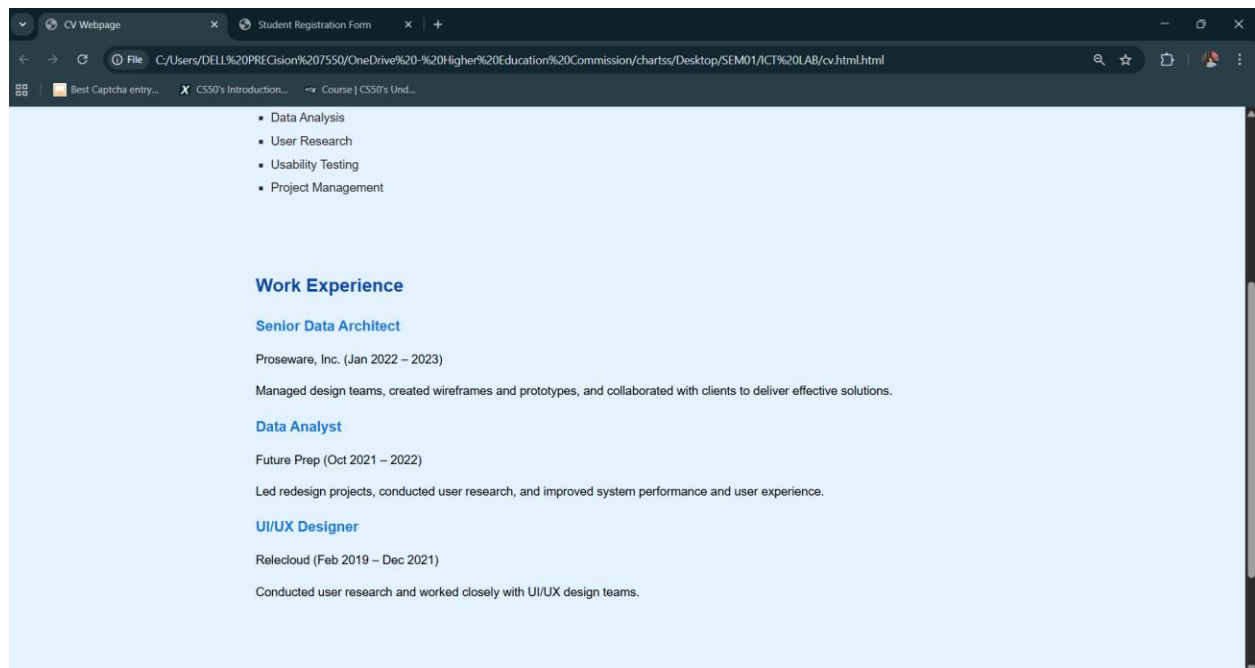
- High contrast color scheme for better readability.
- Keyboard navigation support for all interactive elements.
- Visible focus indicators on focusable elements.

Screenshots

Where to Add Screenshots

CV webpage:





Registration form:

CV Webpage x Student Registration Form x +

C:/Users/DELL%20PRECision%207550/OneDrive%20-%20Higher%20Education%20Commission/chartss/Desktop/SEM01/ICT%20LAB/reg%20form.html.html

Best Captcha entry... X CSS0's Introduction... -> Course | CSS0's Und...

Home | Register | Contact

Student Registration Form

Instruction	Details
Fill all fields	Mandatory

Personal Details

Name:

Email:

Gender: ☐ Male ☒ Female

Contact Number:

City:

Education Details

CV Webpage x Student Registration Form x +

C:/Users/DELL%20PRECision%207550/OneDrive%20-%20Higher%20Education%20Commission/chartss/Desktop/SEM01/ICT%20LAB/reg%20form.html.html

Best Captcha entry... X CSS0's Introduction... -> Course | CSS0's Und...

Name:

Email:

Gender: ☐ Male ☒ Female

Contact Number:

City:

Education Details

Program:

Skills:

- ☒ Data Analysis
- ☒ User Research
- ☐ HTML
- ☐ CSS

© 2025 Student Registration System

Code Analysis & Improvements

Strengths of Current Implementation

- Clean separation of HTML structure and CSS styling.
- Consistent color scheme and typography.
- Good responsive design approach.
- Effective use of modern CSS3 features such as gradients and transitions.

Suggested Enhancements

Animation for page load:

css

```
@keyframes fadeIn {  
    from { opacity: 0; transform: translateY(20px); }  
    to { opacity: 1; transform: translateY(0); }  
}
```

```
section {  
    animation: fadeIn 0.6s ease-out;  
}
```

Print styles:

css

```
@media print {  
    nav,  
    footer {  
        display: none;  
    }  
}
```

```
body {  
    font-size: 12pt;  
    color: black;  
    background: white;  
}
```

```
section {  
    box-shadow: none;  
    border: 1px solid #ddd;  
}
```

```
}
```

Dark mode support:

css

```
@media (prefers-color-scheme: dark) {
```

```
    body {  
        background-color: #121212;  
        color: #ffffff;  
    }
```

```
    section {  
        background-color: #1e1e1e;  
        color: #ffffff;  
    }
```

}

Challenges Faced & Solutions

Challenge 1: Cross-Browser Compatibility

- Problem: CSS Grid not fully supported in some older browsers.
- Solution: Implemented flexbox as a fallback layout and used feature detection.

Challenge 2: Form Validation Styling

- Problem: Browser default validation messages were inconsistent.
- Solution: Implemented custom validation using JavaScript and CSS classes for error/success states.

Challenge 3: Responsive Navigation

- Problem: Navigation bar layout broke on small screens.
 - Solution: Implemented a hamburger menu for mobile view using CSS and JavaScript toggle logic.
-

Conclusion

This lab demonstrated advanced CSS styling techniques for building professional web interfaces. The CV webpage implements modern design principles including responsive layout, interactive components, and accessibility considerations. The registration form applies user-friendly patterns with visual feedback and validation to improve the overall user experience.

Key achievements include:

- Mastery of CSS selectors and specificity rules.
- Implementation of modern CSS3 features such as gradients, transitions, and shadows.
- Creation of responsive, mobile-friendly layouts.

- Understanding cross-browser compatibility issues and strategies.
 - Application of basic web accessibility standards in layout and interaction.
-

References

- MDN Web Docs – CSS Documentation.
- W3Schools – CSS Tutorials.
- CSS-Tricks – Advanced CSS Techniques.
- Google Material Design Guidelines.
- Web Accessibility Initiative (WAI) Guidelines

